

Diabetes Prediction System

Overview

The Diabetes Prediction System is a Machine Learning web application that predicts whether a patient is likely to have diabetes based on several health-related parameters.

The application uses a trained Support Vector Machine (SVM) model and provides predictions through an interactive Streamlit web interface.

Features

- Predicts diabetes risk in real-time
 - User-friendly web interface
 - Machine Learning model trained using Scikit-Learn
 - Deployable on Streamlit Community Cloud
 - Fast and lightweight application
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Input Parameters

The model uses the following medical attributes:

1. Pregnancies
 2. Glucose Level
 3. Blood Pressure
 4. Skin Thickness
 5. Insulin
 6. BMI (Body Mass Index)
 7. Diabetes Pedigree Function
 8. Age
-

Technologies Used

- Python
 - Streamlit
 - Scikit-Learn
 - NumPy
 - Joblib
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Project Structure

```
Diabetes-Prediction/  
|  
├─ app.py  
├─ diabeties.pkl  
├─ requirements.txt  
├─ README.md  
└─ assets/
```

Installation

Clone the repository:

```
git clone <repository-url>  
cd Diabetes-Prediction
```

Install dependencies:

```
pip install -r requirements.txt
```

Run the Application

```
streamlit run app.py
```

The application will open automatically in your browser.

Sample Test Input (Likely Non-Diabetic)

Feature	Value
Pregnancies	1
Glucose	95
Blood Pressure	70
Skin Thickness	20
Insulin	85

Feature	Value
BMI	23.5
Diabetes Pedigree Function	0.25
Age	25

Machine Learning Model

Algorithm Used:

- Support Vector Machine (SVM)

Target Classes:

- 0 → Non-Diabetic
- 1 → Diabetic

Future Improvements

- Probability score display
- Model confidence visualization
- Health recommendation system
- Patient history tracking
- Enhanced dashboard analytics

Author

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